

Fig. 3 (a): Typical boost switching regulator; (b) typical boost topology maximum power point converter

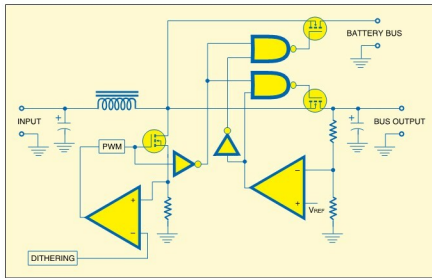


Fig. 4: Dual-output MPP module

hours in eclipse, our efficiency boosts back up to 69%—the efficiency of the original simpler system.

MOSFET switches the inductor to ground, building a current in the inductor. When it reaches the desired

Sixty-nine per cent may not sound bad; but remember, we are still throwing away a watt for every 2.2 watts we receive from the solar array. The question is, can we do any better by further rearranging the topology? The answer is yes, if we are willing to break from the traditional blocks and think a little outside the box.

One optimisation we can try is to modify the MPP to generate two outputs, one that is a regulated output voltage for bus voltage, and the other which is a current output for charging the battery. This modification is actually fairly simple to do, but requires a little background on the operation of the MPP converter.

Let's start with a typical boost-topology switching regulator circuit, like the one in Fig. 3(a). A

level, the MOSFET turns off. The current in the inductor then forward biases the diode, which channels the inductive kick-back into the output capacitor. The output voltage is then sampled, filtered and fed to the PWM generator, to determine the amount of inductor current that must be generated by the next pulse.

An MPP converter is basically the same circuit, except that the inductor current is set by the dithering circuit, not the feedback.

The feedback is used, instead, to determine whether the next pulse should be generated, not its magnitude. The result is a sharing of pulse control by the dithering circuit and the feedback control. (See Fig. 3(b).)

To increase efficiency, the diode is often replaced with a second MOSFET, which turns on during the off time of the first MOSFET. This reduces the losses by shorting the 0.5V drop of the diode, with the 10-20 milliohms resistance of a MOSFET.

To create our dual-output MPP, all we need to do is replace the diode with two MOSFETs: one that drives the capacitor connected to the load's switching regulator, and a second that drives a current bus for charging the batteries. The resulting circuit is a dual-output MPP, which simply routes every current pulse where it is needed (the output or the battery). If the constant voltage output is low, as determined by the feedback, the first MOSFET routes the inductive kick to the output capacitor to shore up the voltage. If the constant output voltage is high, the second MOSFET is used to route the inductive kick to the batteries for charging. (See Fig. 4.)

This modification would leave our daytime efficiency at 72%, but raises our night-time efficiency, because we won't need a separate boost regulator to charge the battery. The result is an increase to 65% for the night-time efficiency:

$$\text{System eff. night} = (\text{MPP eff.}) \times (\text{battery charge eff.}) \times (\text{switcher} - \text{diode loss})$$